

Introduction to Programming

COMP101

2025-26

Lab-07

Objectives:

- Exercise-01 Nested construct while-for-if
- Exercise-02 Validating user input with while not in
- Exercise-03 Validating user input from a sequence
- Exercise-04 Validating user input from a range

OR

Work assignment

Exercise-01:

Nested while – for – if

In assignment-04, without validation, the overall structure could look like:

```
input Are there assignments to mark (Y/N)           #assume Y is input

while input = 'Y':
    input number_of_elements (1, 2 or 3)           # works for 1, 2, 3

    for i in range(number_of_elements): #for loop nested in while loop
        output ('Element', i)           #i tells us the element number
        input a grade
        input number of days late (0, 1, 2, 3, 4, 5)

        if late == 0:                     # if-elif-else nested in for loop
            final grade = grade minus 0
            output final grade

        elif late = 1:
            final grade = grade minus 5
            output final grade

        elif late = 2:
            final grade = grade minus 10
            output final grade

        else:                             #else captures the exception (3, 4, 5)
            final grade = Fail

    input Are there assignments to mark (Y/N)
    #note it lines up with the for loop above. We have exited the for loop and
    are back in the while loop (the statement is indented under the while
    statement).

    When Y is input, we process the next student. When N is input we exit
    the while loop and end the program

Output end-of-program  #while loop is finished
```

Exercise-02: Validating user input with while not in

This can be used to make a user enter what you want them to enter.
Cut & paste into IDLE and test it with input that is not 'Y' and not 'N'.

User is asked if they want to input grades – the correct response is only Y or N

The while loop will keep asking for Y or N to be input until user enters the correct response (Y, or N)

It works for any invalid input including strings and numbers. Try it.

```
assignment = input('Are there assignments to grade (Y/N):')
assignment = assignment.upper()

while assignment not in ('Y','N'):
    print ('Invalid - enter Y or N')
    assignment = input('Any there any assignments to grade (Y/N): ')
    assignment = assignment.upper()

print ('Input ok')
```

Exercise-03: Validating user input from a sequence

This can be used to make a user enter what you want them to enter.
Cut & paste into IDLE and test it.

User is asked to input a number from a sequence:
The correct response is only the numbers in the sequence.

Given: (2, 4, 6, 8) means we can only use these numbers

The while loop will keep asking for a number in the sequence until user enters the correct response (2 or 4 or 6 or 8)

It works for any number but not for strings.

It works for any invalid number but not for strings (because we ask for int input and strings can't assign to int).

We would have to remove the 'int' and then when a number is input (as a string) we can cast it to int.

```
evens = int(input('Enter 2 or 4 or 6 or 8 '))
```

```
while evens not in (2, 4, 6, 8):
```

```
    print ('invalid')
```

```
    evens = int(input('Enter 2 or 4 or 6 or 8 '))
```

```
print ('Correct input - end of program')
```

Exercise-04: Validating user input from a range

This can be used to make a user enter what you want them to enter.
Cut & paste into IDLE and test it.

User is asked to input a number from a range:
The correct response is only the numbers in the range

Given: range 1 to 50 – means we can input any number in between

The while loop will keep asking for a number in the range of 1 to 50 inclusive when the user inputs a number outside the range boundaries.

It works for numbers but not for strings.

It works for any invalid number but not for strings (because we ask for int input and strings can't assign to int).

We would have to remove the 'int' and then when a number is input (as a string) we can cast it to int.

```
num = int(input('Enter a number between 1 and 50 '))
```

```
while num not in range (1,51):
```

```
    print ('invalid')
```

```
    num = int(input('Enter a number between 1 and 50 '))
```

```
print ('Correct input - end of program')
```

Note the range (1,51) to allow for the range index starting at [0]